

Applications of Derivatives

L'Hôpital, Polynomial Roots, and Inequalities

A session on derivative-based techniques across analysis and algebra

Section 1

The L'Hôpital Toolkit

L'Hôpital's Rule: the general idea

Concept

L'Hôpital's rule applies to indeterminate forms of type

$$\frac{0}{0} \quad \text{and} \quad \frac{\bullet}{\infty},$$

where “ $\bullet$ ” is *anything at all*: a finite limit, $\pm\infty$, or even an undefined oscillating quantity. The numerator's behavior is irrelevant when the denominator blows up.

The key insight: if the denominator escapes to infinity, the rule fires regardless of what the numerator is doing.

L'Hôpital and its discrete cousin

L'Hôpital (general form)

Let f, g be differentiable for sufficiently large x , with $g'(x) \neq 0$. If

$$\lim_{x \rightarrow \infty} g(x) = \pm\infty$$

and

$$\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = L \in \overline{\mathbb{R}},$$

then

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L.$$

Stolz–Cesàro (discrete)

Let $(a_n), (b_n)$ be real sequences with (b_n) strictly monotonic and

$$\lim_{n \rightarrow \infty} b_n = \pm\infty.$$

If

$$\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = L \in \overline{\mathbb{R}},$$

then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L.$$

Differences replace derivatives; the conclusion is identical.

Application 1: A classic limit

Problem

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable, and suppose

$$\lim_{x \rightarrow \infty} (f(x) + f'(x)) = L$$

for some finite $L \in \mathbb{R}$. Show that $\lim_{x \rightarrow \infty} f(x) = L$.

*Strategy: Multiply numerator and denominator by e^x
to expose an ∞/∞ form, then apply L'Hôpital.*

Application 1 (*solution via Hardy's trick*)

The trick. Write

$$f(x) = \frac{e^x f(x)}{e^x}.$$

The denominator $e^x \rightarrow \infty$, so this is an $\bullet/\infty$ form, and L'Hôpital applies regardless of whether the numerator has a limit.

Apply L'Hôpital.

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{e^x f(x)}{e^x} = \lim_{x \rightarrow \infty} \frac{e^x f(x) + e^x f'(x)}{e^x} = \lim_{x \rightarrow \infty} (f(x) + f'(x)) = L$$



The hypothesis “ $\lim(f + f')$ exists” becomes the limit of a derivative quotient, which L'Hôpital identifies with the limit of f itself.

Application 1 (*alternative solution via MVT*)

Setup. Define $g(x) = e^x f(x)$. Then $g'(x) = e^x (f(x) + f'(x))$.

Apply LMVT on $[x, x+1]$. There exists $c_x \in (x, x+1)$ with

$$g(x+1) - g(x) = g'(c_x) = e^{c_x} (f(c_x) + f'(c_x)).$$

That is,

$$e^{x+1} f(x+1) - e^x f(x) = e^{c_x} (f(c_x) + f'(c_x)).$$

Rearrange.

$$\frac{e^{x+1} f(x+1) - e^x f(x)}{e^{x+1} - e^x} = \frac{e^{c_x}}{e^{x+1} - e^x} (f(c_x) + f'(c_x)).$$

As $x \rightarrow \infty$, $c_x \rightarrow \infty$ as well, so $f(c_x) + f'(c_x) \rightarrow L$. The discrete Stolz–Cesàro form (with $a_n = e^n f(n)$, $b_n = e^n$) then gives $\lim f(x) = L$. ■

MCQ: A weighted variant

Problem

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable with

$$\lim_{x \rightarrow \infty} (3f(x) + 4f'(x)) = L, \quad L \in \mathbb{R}.$$

Then $\lim_{x \rightarrow \infty} f(x)$ equals:

(A) $3L$

(B) $L/3$

(C) $4L$

(D) $L/4$

Hint: choose the right exponential multiplier.

Answer: (B) $L/3$.

Hardy's trick. Seek a multiplier $e^{\alpha x}$ so that

$$\frac{d}{dx}(e^{\alpha x} f(x)) = e^{\alpha x}(\alpha f(x) + f'(x))$$

matches $3f + 4f'$ up to a constant. Matching ratios $\alpha/1 = 3/4$ gives $\alpha = 3/4$.

Apply L'Hôpital. Writing $f(x) = \frac{e^{(3/4)x} f(x)}{e^{(3/4)x}}$,

$$\begin{aligned} \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{e^{(3/4)x} \left(\frac{3}{4} f(x) + f'(x) \right)}{\frac{3}{4} e^{(3/4)x}} = \lim_{x \rightarrow \infty} \frac{\frac{3}{4} f(x) + f'(x)}{\frac{3}{4}} \\ &= \frac{1}{3} \lim_{x \rightarrow \infty} (3f(x) + 4f'(x)) = \frac{L}{3}. \end{aligned}$$



The general $af + bf'$ rule

Generalization

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable, let $a, b > 0$, and suppose

$$\lim_{x \rightarrow \infty} (a f(x) + b f'(x)) = L \in \mathbb{R}.$$

Then $\lim_{x \rightarrow \infty} f(x) = \frac{L}{a}.$

Proof sketch. Apply Hardy's trick with multiplier $e^{(a/b)x}$:

$$\frac{d}{dx} (e^{(a/b)x} f(x)) = \frac{1}{b} e^{(a/b)x} (a f(x) + b f'(x)).$$

L'Hôpital on $f(x) = e^{(a/b)x} f(x) / e^{(a/b)x}$ gives $\lim f(x) = \lim \frac{a f(x) + b f'(x)}{a} = \frac{L}{a}.$

Note: The condition $a, b > 0$ ensures the multiplier $e^{(a/b)x} \rightarrow \infty$.

MCQ: A higher-order trap

Problem

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be thrice differentiable with

$$\lim_{x \rightarrow \infty} (f(x) + f'(x) + f''(x) + f'''(x)) = L, \quad L \in \mathbb{R}.$$

Then $\lim_{x \rightarrow \infty} f(x)$ equals:

- (A) L
- (B) $L/2$
- (C) $L/4$
- (D) Cannot be determined

Hint: try a function whose derivatives cycle.

Answer: (D) Cannot be determined.

Counterexample. Take $f(x) = \sin x$. Then

$$f'(x) = \cos x, \quad f''(x) = -\sin x, \quad f'''(x) = -\cos x,$$

so

$$f(x) + f'(x) + f''(x) + f'''(x) = \sin x + \cos x - \sin x - \cos x = 0.$$

Hence $\lim_{x \rightarrow \infty} (f + f' + f'' + f''') = 0$, satisfying the condition with $L = 0$. But $\lim_{x \rightarrow \infty} \sin x$ does not exist! ■

Section 2

Derivatives in Polynomial Algebra

The log-derivative principle

Key identity

If $P(x) = \prod_{i=1}^n (x - a_i)$, then taking logs and differentiating gives

$$\frac{P'(x)}{P(x)} = \sum_{i=1}^n \frac{1}{x - a_i}.$$

Differentiating once more:

$$\frac{(P')^2 - P''P}{P^2} = \sum_{i=1}^n \frac{1}{(x - a_i)^2}.$$

This converts symmetric sums over roots into evaluations of P, P', P'' at convenient points.

Application 1: A 50-term sum

Problem

Let $a_1, a_2, \dots, a_{50}$ be the roots of

$$1 + x + x^2 + \cdots + x^{50} = 0.$$

Find $\sum_{i=1}^{50} \frac{1}{(1 - a_i)^2}$.

Strategy: Use the log-derivative formula and evaluate P, P', P'' at $x = 1$.

Application 1 (*solution*)

Setup. Let $P(x) = 1 + x + x^2 + \cdots + x^{50}$. Then $P(x) = \prod_{i=1}^{50} (x - a_i)$ and

$$\sum_{i=1}^{50} \frac{1}{(1 - a_i)^2} = \sum_{i=1}^{50} \frac{1}{(a_i - 1)^2} = \left. \frac{(P'(x))^2 - P''(x)P(x)}{P(x)^2} \right|_{x=1}.$$

Compute the values at $x = 1$.

$$P(1) = 51,$$

$$P'(1) = 1 + 2 + 3 + \cdots + 50 = \frac{50 \cdot 51}{2} = 1275,$$

$$P''(1) = \sum_{k=2}^{50} k(k-1) = \frac{50 \cdot 51 \cdot 101}{6} - \frac{50 \cdot 51}{2} = 42925 - 1275 = 41650.$$

Application 1 (*continued*)

Plug in.

$$\sum_{i=1}^{50} \frac{1}{(1 - a_i)^2} = \frac{1275^2 - 51 \cdot 41650}{51^2} = \frac{-498\,525}{2601} = \boxed{-\frac{575}{3}}.$$

(using $\gcd(498525, 2601) = 867$.)



The a_i are non-real 51st roots of unity, so individual terms $1/(1 - a_i)^2$ are complex; their sum is real but need not be positive.

Application 2: A first-order sum

Problem

Let $a_1, a_2, \dots, a_{n-1}$ be the n -th roots of unity other than 1 (i.e., the roots of $1 + x + x^2 + \dots + x^{n-1} = 0$). Find a closed form for

$$\sum_{i=1}^{n-1} \frac{1}{1 - a_i}.$$

Strategy: Apply the first-order log-derivative formula at $x = 1$.

Application 2 (*solution*)

Setup. Let $Q(x) = 1 + x + \cdots + x^{n-1} = \prod_{i=1}^{n-1} (x - a_i)$. By log-differentiation,

$$\frac{Q'(x)}{Q(x)} = \sum_{i=1}^{n-1} \frac{1}{x - a_i}, \quad \text{so} \quad \sum_{i=1}^{n-1} \frac{1}{1 - a_i} = \frac{Q'(1)}{Q(1)}.$$

Compute. $Q(1) = n$ and $Q'(1) = 1 + 2 + \cdots + (n-1) = \frac{n(n-1)}{2}$. Therefore

$$\sum_{i=1}^{n-1} \frac{1}{1 - a_i} = \frac{n(n-1)/2}{n} = \boxed{\frac{n-1}{2}}.$$



Application 3: A weighted sum at $x = 2$

Problem

Let $a_1, a_2, \dots, a_n$ be the n -th roots of unity (i.e., the roots of $x^n - 1 = 0$). Find

$$\sum_{i=1}^n \frac{a_i}{2 - a_i}.$$

Strategy: Decompose $\frac{a_i}{2 - a_i} = -1 + \frac{2}{2 - a_i}$, then apply log-differentiation at $x = 2$.

Application 3 (solution)

Decomposition.

$$\frac{a_i}{2 - a_i} = \frac{-(2 - a_i) + 2}{2 - a_i} = -1 + \frac{2}{2 - a_i}.$$

Summing over $i = 1, \dots, n$:

$$\sum_{i=1}^n \frac{a_i}{2 - a_i} = -n + 2 \sum_{i=1}^n \frac{1}{2 - a_i}.$$

Log-derivative at $x = 2$. With $P(x) = x^n - 1 = \prod_{i=1}^n (x - a_i)$,

$$\sum_{i=1}^n \frac{1}{2 - a_i} = \frac{P'(2)}{P(2)} = \frac{n \cdot 2^{n-1}}{2^n - 1}.$$

Application 3 (*continued*)

Combine.

$$\begin{aligned}\sum_{i=1}^n \frac{a_i}{2 - a_i} &= -n + 2 \cdot \frac{n \cdot 2^{n-1}}{2^n - 1} = -n + \frac{n \cdot 2^n}{2^n - 1} \\ &= \frac{-n(2^n - 1) + n \cdot 2^n}{2^n - 1} = \boxed{\frac{n}{2^n - 1}}.\end{aligned}$$



The key step: split the awkward $a_i/(2 - a_i)$ into a constant plus a clean reciprocal that the log-derivative formula handles.

Application 4: A vanishing sum identity

Problem

Let $P(x) = (x-1)(x-2)\cdots(x-n)$ where $n \geq 2$. Show that

$$\sum_{k=1}^n \frac{1}{P'(k)} = 0.$$

Strategy: Use partial-fraction decomposition of $1/P(x)$, then take $x \rightarrow \infty$.

Application 4 (*solution*)

Partial fractions. Since P has n distinct simple roots at $1, 2, \dots, n$,

$$\frac{1}{P(x)} = \sum_{k=1}^n \frac{1}{P'(k)(x-k)}.$$

This is the standard residue formula: the coefficient at the simple pole $x = k$ is $1/P'(k)$.

Multiply by x .

$$\frac{x}{P(x)} = \sum_{k=1}^n \frac{1}{P'(k)} \cdot \frac{x}{x-k}.$$

Application 4 (*continued*)

Take $x \rightarrow \infty$.

► **LHS:** since $\deg P = n \geq 2$, $\frac{x}{P(x)} \rightarrow 0$.

► **RHS:** $\frac{x}{x-k} \rightarrow 1$ for each fixed k , so the sum tends to $\sum_{k=1}^n \frac{1}{P'(k)}$.

Equating limits:

$$0 = \sum_{k=1}^n \frac{1}{P'(k)}.$$



The hypothesis $n \geq 2$ is essential — for $n = 1$, $P(x) = x - 1$ and $1/P'(1) = 1 \neq 0$.

Section 3

Using Derivatives to Locate Roots

Application 1: An exponential equation

Problem

Show that the equation

$$3^x + 4^x = 5^x$$

has exactly one real solution.

*Strategy: Divide through by 5^x to obtain a strictly monotonic function,
then apply IVT and the monotonicity criterion.*

Application 1 (*solution*)

Reformulate. Divide both sides by $5^x > 0$:

$$3^x + 4^x = 5^x \iff \left(\frac{3}{5}\right)^x + \left(\frac{4}{5}\right)^x = 1.$$

Let $h(x) = (3/5)^x + (4/5)^x$. We want to show $h(x) = 1$ has exactly one solution.

Monotonicity via the derivative. Since $0 < 3/5, 4/5 < 1$,

$$h'(x) = \ln\left(\frac{3}{5}\right) \cdot \left(\frac{3}{5}\right)^x + \ln\left(\frac{4}{5}\right) \cdot \left(\frac{4}{5}\right)^x < 0 \quad \forall x \in \mathbb{R},$$

since both logarithms are negative. So h is *strictly decreasing* on $\mathbb{R}$.

Boundary behavior.

$$\lim_{x \rightarrow -\infty} h(x) = +\infty, \quad \lim_{x \rightarrow +\infty} h(x) = 0.$$

By IVT, h takes the value 1 exactly once. (And one easily checks $x = 2$ is the solution: $9 + 16 = 25$.) ■

Application 2: A logarithmic equation

Problem

Show that the equation

$$x \ln x = 1$$

has exactly one real solution in $(0, \infty)$.

Strategy: Analyze the function $f(x) = x \ln x$ via its derivative to find its minimum and monotonicity intervals.

Application 2 (*solution*)

Set up. Let $f(x) = x \ln x$ on $(0, \infty)$. Compute

$$f'(x) = \ln x + 1, \quad f'(x) = 0 \iff x = \frac{1}{e}.$$

So f is strictly decreasing on $(0, 1/e)$ and strictly increasing on $(1/e, \infty)$, with minimum value

$$f(1/e) = \frac{1}{e} \ln \frac{1}{e} = -\frac{1}{e} < 0.$$

Behavior on each branch.

- ▶ On $(0, 1/e]$: f is strictly decreasing and $f(x) \rightarrow 0^-$ as $x \rightarrow 0^+$. So $f(x) \leq 0$ throughout, never reaching 1.
- ▶ On $[1/e, \infty)$: f is strictly increasing from $-1/e$ at $x = 1/e$ to $+\infty$ as $x \rightarrow \infty$. By IVT, f takes the value 1 exactly once on this branch.

Hence $x \ln x = 1$ has exactly one real solution. ■

Section 4

Inequalities via Derivatives

Application 1: Napier's inequality

Problem

Show that for all $0 < a < b$,

$$\frac{1}{b} < \frac{\ln b - \ln a}{b - a} < \frac{1}{a}.$$

*Strategy: Apply Lagrange's MVT to the function $\ln$ on $[a, b]$,
then bound the derivative $1/x$ on this interval.*

Application 1 (*solution*)

Apply LMVT to $\ln$ on $[a, b]$. Since $\ln$ is continuous on $[a, b]$ and differentiable on (a, b) , there exists $c \in (a, b)$ with

$$\frac{\ln b - \ln a}{b - a} = (\ln)'(c) = \frac{1}{c}.$$

Bound $1/c$. Since $a < c < b$ and $x \mapsto 1/x$ is strictly decreasing on $(0, \infty)$,

$$\frac{1}{b} < \frac{1}{c} < \frac{1}{a}.$$

Substituting back gives Napier's inequality $\frac{1}{b} < \frac{\ln b - \ln a}{b - a} < \frac{1}{a}$. ■

This sandwiches the logarithmic mean $\frac{b - a}{\ln b - \ln a}$ between a and b .

Application 2: A second-order log estimate

Problem

Show that for all $x > 1$,

$$\ln x > \frac{2(x-1)}{x+1}.$$

Strategy: Form the difference, check the value at $x = 1$, and look for an algebraic factorization in the derivative.

Application 2 (solution)

Setup. Define $f(x) = \ln x - \frac{2(x-1)}{x+1}$ on $[1, \infty)$. Then $f(1) = 0$.

Compute the derivative.

$$f'(x) = \frac{1}{x} - \frac{2(x+1) - 2(x-1)}{(x+1)^2} = \frac{1}{x} - \frac{4}{(x+1)^2} = \frac{(x+1)^2 - 4x}{x(x+1)^2} = \frac{(x-1)^2}{x(x+1)^2}$$

Conclude. For $x > 1$, $f'(x) > 0$, so f is strictly increasing on $[1, \infty)$. Combined with $f(1) = 0$,

$$f(x) > 0 \implies \ln x > \frac{2(x-1)}{x+1} \quad \forall x > 1.$$



The factorization $(x+1)^2 - 4x = (x-1)^2$ is the key — it forces $f' \geq 0$ algebraically.

Application 3: Cusa–Huygens inequality

Problem

Show that for all $x \in (0, \pi/2)$,

$$\frac{\sin x}{x} < \frac{2 + \cos x}{3}.$$

Strategy: Clear denominators and apply the iterated derivative recipe three times — until the third derivative reveals a clear sign.

Application 3 (*solution*)

Reformulate. The inequality is equivalent to

$$f(x) := (2 + \cos x)x - 3\sin x > 0 \quad \text{on } (0, \pi/2).$$

Note $f(0) = 0$. We will show f', f'', f''' all vanish at 0 and $f''' > 0$ on $(0, \pi/2)$.

Compute the derivatives.

$$\begin{aligned} f'(x) &= (2 + \cos x) - x \sin x - 3 \cos x = 2 - 2 \cos x - x \sin x, \\ f''(x) &= 2 \sin x - \sin x - x \cos x = \sin x - x \cos x, \\ f'''(x) &= \cos x - \cos x + x \sin x = x \sin x. \end{aligned}$$

At $x = 0$: $f'(0) = f''(0) = f'''(0) = 0$ (and $f(0) = 0$).

Application 3 (*continued*)

Cascade upward. On $(0, \pi/2)$, $f'''(x) = x \sin x > 0$ (both factors positive).

Hence f'' is strictly increasing on $[0, \pi/2)$. Combined with $f''(0) = 0$:

$$f''(x) > 0 \quad \forall x \in (0, \pi/2).$$

Hence f' is strictly increasing on $[0, \pi/2)$. Combined with $f'(0) = 0$:

$$f'(x) > 0 \quad \forall x \in (0, \pi/2).$$

Hence f is strictly increasing on $[0, \pi/2)$. Combined with $f(0) = 0$:

$$f(x) = (2 + \cos x)x - 3 \sin x > 0,$$

which rearranges to $\frac{\sin x}{x} < \frac{2 + \cos x}{3}$. ■

The recipe “ $f(0) = 0, f' > 0 \Rightarrow f > 0$ ” iterated three times.

Thank You

Questions?